Niagara REST API

User Guide & Test Samples

Module: **niagaraRestApi** v1.0.1.4.14

Station: openClawTestRestful (N4.14.0.162) @ 192.168.2.97:8081

Vendor: Shanghai Gline Net Co.

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1. Overview & Architecture

The niagaraRestApi module is a custom Niagara N4 module that embeds an HTTP server inside a Niagara Station. It exposes a JSON-based REST API for programmatic interaction with the station - all without requiring SMA (Software Maintenance Agreement) or OEM licenses.

Architecture

```
BRestApiService (extends BComponent)
+-- enabled (boolean)
+-- port (int, default 8081)
+-- started() -> Creates HttpServer
+-- stopped() -> Stops HttpServer

Handlers:
  Status  -> GET  /api/status
  Modules -> GET  /api/modules
  Types   -> GET  /api/types?module=xxx
  Config  -> GET  /api/config?path=/xxx
  Batch   -> POST /api/batch
```

• Uses JDK built-in HttpServer - no external web container.

• JSON-only request/response format.

• **No built-in authentication** - designed for internal LAN only.

• Batch mutation model: all writes go through POST /api/batch.

• Station: openClawTestRestful (N4.14.0.162) @ 192.168.2.97:8081.

2. Authentication

No Built-in Authentication

The niagaraRestApi module has no authentication. It is designed for internal LAN use only. For production deployments, add protection via firewall ACLs or a reverse proxy (nginx/Apache) with HTTP Basic Auth.

■ Do NOT expose port 8081 to the public internet without additional protection.

3. REST API Endpoints Reference

3.1 GET /api/status

Check if the REST API service is running.

```
curl http://192.168.2.97:8081/api/status
{"status":"ok","name":"niagaraRestApi","port":8081}
```

3.2 GET /api/modules

List all loaded modules with type counts.

```
curl http://192.168.2.97:8081/api/modules
[{"name":"baja","types":347},{"name":"modbusCore","types":77},{"name":"modbusTcp","types":7},{"name":"niagaraRestApi","types
```

3.3 GET /api/types

Get all component types for a given module.

```
curl "http://192.168.2.97:8081/api/types?module=control"
[{"type":"control:NumericWritable","module":"control"},{"type":"control:BooleanWritable","module":"control"},{"type":"contro
```

Common Module Types

Module	Common Types	Description
baja	Folder	Container component
control	NumericWritable, BooleanWritable, EnumWritable	Point types
modbusTcp	ModbusTcpNetwork, ModbusTcpDevice	TCP network/device
modbusCore	NumericProxyExt, BooleanProxyExt	Modbus proxy extension
kitControl	Add, Ramp, SineWave, Random	Simulation signals

3.4 GET /api/config

Browse the station configuration tree at any path.

```
curl "http://192.168.2.97:8081/api/config?path=/"
{"name":"Station","type":"Station","children":[{"name":"AlarmService","type":"AlarmService"},{"name":"Drivers","type":"Drive
```

3.5 POST /api/batch - Core Operation

All mutations go through this endpoint. Send a JSON array of operation objects executed sequentially in order.

Common Fields

Field	Type	Description	Applies To
OP	string	Operation type	ALL
PATH	string	Component path from /	ALL
T	string	Type: module:TypeName	add
N	string	Component/slot name	add,mod,link
NEW	string	New name	rename,cp
V	any	Value (strings need quotes)	mod
TARGET	string	Target path	link,unlink
TARGET_N	string	Target slot	link
VAL	any	Invoke value	invoke
VAL_TYPE	string	num/bool/str	invoke
TARGET_PATH	string	Copy destination	cp

10 Operations

1. ADD - Create

Creates a child component under the parent path.

```
{"OP":"add","PATH":"/","T":"baja:Folder","N":"myFolder"}
```

2. LINK - Connect

Connects source.out to target.in10 via BConversionLink with auto-type-conversion.

```
{"OP":"link","PATH":"/src","N":"out","TARGET":"/tgt","TARGET_N":"in10"}
```

3. RENAME - Rename

Renames using parent.rename() - preserves all existing links (handle-based ORD).

```
{"OP":"rename","PATH":"/folder/old","NEW":"new"}
```

4. CP - Deep Copy

Recursive deep copy of entire subtree. Internal links rebuilt in copy.

```
{"OP":"cp","PATH":"/folder","NEW":"copy","TARGET_PATH":"/"}
```

5. MOD - Modify Value

Writes a property value. Strings need quotes; numbers bare; booleans true/false.

```
{"OP":"mod","PATH":"/device","N":"ipAddress","V":"192.168.2.140"}
```

6. UNLINK - Disconnect

Removes the Link child from the target point.

```
{"OP":"unlink","TARGET":"/tgtPoint"}
```

7. REMOVE - Delete

Permanently removes the component from the station tree.

```
{"OP":"remove","PATH":"/parent/child"}
```

8. READ - Read Property

Returns current value. Omit N to get full component info (type, slots, children).

```
{"OP":"read","PATH":"/device","N":"ipAddress"}
```

9. LIST - List Children

Returns names and types of all immediate child components.

```
{"OP":"list","PATH":"/myFolder"}
```

10. INVOKE - Call Action

Invokes a Niagara action. VAL_TYPE supports: num, bool, str.

```
{"OP":"invoke","PATH":"/point/set","VAL":"42","VAL_TYPE":"num"}
```

MOD Special Values

Type	JSON Format	Example
String	V:"value"	`V:"192.168.2.140"`
Integer	V:number	`V:502`
Boolean	V:true/false	`V:true`
FlexAddress	V:"addr|format"	`V:"2|decimal"`

- Enum types (regType, dataType) cannot be written via MOD (BString-to-enum conversion fails).
- BFacets types (deviceFacets) cannot be written via MOD.

4. Verified Test Samples

All operations tested on station 192.168.2.97:8081 (openClawTestRestful) with admin/admin credentials.

4.1 Create Modbus TCP Device (T8600 Thermostat)

Creates a Modbus TCP device with 3 points (OnOff, Mode, SetTemp), configures register addresses, sets IP 192.168.2.140:502, and pings to verify connectivity.

```
curl -X POST http://192.168.2.97:8081/api/batch -H "Content-Type: application/json" -d '[
{"OP":"add","PATH":"/Drivers/ModbusTcpNetwork","T":"modbusTcp:ModbusTcpDevice","N":"T8600_new"},
{"OP":"mod","PATH":"/.../T8600_new","N":"deviceAddress","V":2},
{"OP":"mod","PATH":"/.../T8600_new","N":"ipAddress","V":"192.168.2.140"},
{"OP":"add","PATH":"/.../T8600_new/points","T":"control:NumericWritable","N":"OnOff"},
{"OP":"add","PATH":"/.../OnOff","T":"modbusCore:ModbusClientNumericProxyExt","N":"proxyExt"},
{"OP":"mod","PATH":"/.../OnOff/proxyExt","N":"dataAddress","V":"2|decimal"},
{"OP":"add","PATH":"/.../T8600_new/points","T":"control:NumericWritable","N":"Mode"},
{"OP":"add","PATH":"/.../Mode","T":"modbusCore:ModbusClientNumericProxyExt","N":"proxyExt"},
{"OP":"mod","PATH":"/.../Mode/proxyExt","N":"dataAddress","V":"3|decimal"},
{"OP":"add","PATH":"/.../T8600_new/points","T":"control:NumericWritable","N":"SetTemp"},
{"OP":"add","PATH":"/.../SetTemp","T":"modbusCore:ModbusClientNumericProxyExt","N":"proxyExt"},
{"OP":"mod","PATH":"/.../SetTemp/proxyExt","N":"dataAddress","V":"4|decimal"},
{"OP":"invoke","PATH":"/Drivers/ModbusTcpNetwork/T8600_new/ping"}]'
```

4.2 MQTT + BooleanWritable Points

Creates 10 BooleanWritable points under the MQTT folder (tested on 192.168.2.112), then links each to the MqttClient output.

```
[{"OP":"add","PATH":"/MQTT","T":"control:BooleanWritable","N":"Point1"},...,{"OP":"add","PATH":"/MQTT","T":"control:BooleanW
```

4.3 Add + Link + Rename Flow

Test: create folder with two NumericWritable points, link them, then rename the source. Verify the link survives rename (because ORD uses handle h:xxxxx, not name).

```
[{"OP":"add","PATH":"/","T":"baja:Folder","N":"testFolder"},{"OP":"add","PATH":"/testFolder","T":"control:NumericWritable","
Result: renamedSrc still linked to tgtPoint.in10
```

4.4 Deep Copy (CP) + Rename

Create a folder with internal links, deep copy to a new location. Verify internal links are rebuilt in the copy.

```
[{"OP":"add","PATH":"/","T":"baja:Folder","N":"original"},{"OP":"add","PATH":"/original","T":"control:NumericWritable","N":"
Result: copy has copy.point1 link to copy.point2
```

4.5 Add + Link + Unlink + Remove (Full Lifecycle)

Complete component lifecycle: create, link, unlink (disconnect), and delete.

```
[{"OP":"add","PATH":"/","T":"baja:Folder","N":"lifecycle"},{"OP":"add","PATH":"/lifecycle","T":"control:NumericWritable","N"
Result: unlink removes Link child; src deleted; dst remains orphaned
```

5. Python Client Tools

Utility scripts maintained at Y:/claw/skills/115-obix-integration/

5.1 niagara-rest-client.py (9625 bytes)

Universal REST client supporting: login, status, config browse, BQL query, history extraction, point override, and alarm export.

```
python niagara-rest-client.py <host:port> --user admin --password xxx --status
python niagara-rest-client.py <host:port> --user admin --password xxx --batch batch.json
```

5.2 niagara-bql-query.py (6458 bytes)

BQL CLI tool with CSV export, type filtering, and configurable parent path.

```
python niagara-bql-query.py --host 192.168.2.112:80 --user admin --password xxx --query "SELECT * FROM control:BooleanWritab
```

5.3 alarm_to_csv.py (7165 bytes)

Dual-approach alarm export: ProgramService BSH script execution or external Python HTTP client.

```
python alarm_to_csv.py --host 192.168.2.112 --user admin --password xxx --output alarms.csv
```

5.4 Station Exploration Scripts

explore_station.py (2770 bytes): REST API-based station exploration (status, config browse, services).

explore_112.py / explore_112_v2.py: Full SCRAM auth + oBIX XML parsing with tree search into Modbus, BACnet, and simulate folders.

explore_112_obix.py: Python oBIX library-based exploration.

test_connection.py (2085 bytes): Quick connectivity test for 192.168.2.112.

7. Known Limitations

1. No Authentication - internal LAN only. Add reverse proxy for external exposure.
2. Enum Properties (regType, dataType) - cannot write via MOD (BString-to-enum conversion fails).
3. BFacets Properties (deviceFacets) - cannot write via MOD.
4. Module Reload - disabling RestApiService requires full station restart for the module to reload.
5. No Alarm/History Endpoints - REST API has no dedicated alarm or history queries. Use oBIX for those.

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